

# Limin Ge Research Catalog

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## Contents

|                          |   |
|--------------------------|---|
| Real projective geometry | 2 |
| Coding theory            | 2 |
| Discrete geometry        | 4 |

# Real projective geometry

**001. Packings of lines in  $\mathbb{R}^{10}$ .** First values for 7 blank cells ( $N = 101, 103, 106, 109, 112, 119-120$ ) in the row  $m = 10, n = 1$  of the table of best Grassmannian packings, blank in Sloane's 1997 table and in Cohn's current one. Each value beats every packing obtained by deleting subspaces from a larger known one.

$N = 101 \cdot N = 103 \cdot N = 106 \cdot N = 109 \cdot N = 112 \cdot N = 119 \cdot N = 120$

**002. Packings of lines in  $\mathbb{R}^{11}$ .** First values for 17 blank cells ( $N = 84-100$ ) in the row  $m = 11, n = 1$  of the table of best Grassmannian packings, blank in Sloane's 1997 table and in Cohn's current one. Each value beats every packing obtained by deleting subspaces from a larger known one.

$N = 84 \cdot N = 85 \cdot N = 86 \cdot N = 87 \cdot N = 88 \cdot N = 89 \cdot N = 90 \cdot N = 91 \cdot N = 92 \cdot N = 93 \cdot N = 94 \cdot N = 95 \cdot N = 96 \cdot N = 97 \cdot N = 98 \cdot N = 99 \cdot N = 100$

**003. Packings of lines in  $\mathbb{R}^{12}$ .** First values for 14 blank cells ( $N = 95-100, 104-105, 109, 111-112, 116, 119-120$ ) in the row  $m = 12, n = 1$  of the table of best Grassmannian packings, blank in Sloane's 1997 table and in Cohn's current one. Each value beats every packing obtained by deleting subspaces from a larger known one.

$N = 95 \cdot N = 96 \cdot N = 97 \cdot N = 98 \cdot N = 99 \cdot N = 100 \cdot N = 104 \cdot N = 105 \cdot N = 109 \cdot N = 111 \cdot N = 112 \cdot N = 116 \cdot N = 119 \cdot N = 120$

**004. Packings of lines in  $\mathbb{R}^{13}$ .** First values for 44 blank cells ( $N = 55-89, 92-100$ ) in the row  $m = 13, n = 1$  of the table of best Grassmannian packings, blank in Sloane's 1997 table and in Cohn's current one. Each value beats every packing obtained by deleting subspaces from a larger known one.

$N = 55 \cdot N = 56 \cdot N = 57 \cdot N = 58 \cdot N = 59 \cdot N = 60 \cdot N = 61 \cdot N = 62 \cdot N = 63 \cdot N = 64 \cdot N = 65 \cdot N = 66 \cdot N = 67 \cdot N = 68 \cdot N = 69 \cdot N = 70 \cdot N = 71 \cdot N = 72 \cdot N = 73 \cdot N = 74 \cdot N = 75 \cdot N = 76 \cdot N = 77 \cdot N = 78 \cdot N = 79 \cdot N = 80 \cdot N = 81 \cdot N = 82 \cdot N = 83 \cdot N = 84 \cdot N = 85 \cdot N = 86 \cdot N = 87 \cdot N = 88 \cdot N = 89 \cdot N = 92 \cdot N = 93 \cdot N = 94 \cdot N = 95 \cdot N = 96 \cdot N = 97 \cdot N = 98 \cdot N = 99 \cdot N = 100$

**005. Packings of lines in  $\mathbb{R}^{14}$ .** First values for 42 blank cells ( $N = 56-60, 66-100, 105, 120$ ) in the row  $m = 14, n = 1$  of the table of best Grassmannian packings, blank in Sloane's 1997 table and in Cohn's current one. Each value beats every packing obtained by deleting subspaces from a larger known one.

$N = 56 \cdot N = 57 \cdot N = 58 \cdot N = 59 \cdot N = 60 \cdot N = 66 \cdot N = 67 \cdot N = 68 \cdot N = 69 \cdot N = 70 \cdot N = 71 \cdot N = 72 \cdot N = 73 \cdot N = 74 \cdot N = 75 \cdot N = 76 \cdot N = 77 \cdot N = 78 \cdot N = 79 \cdot N = 80 \cdot N = 81 \cdot N = 82 \cdot N = 83 \cdot N = 84 \cdot N = 85 \cdot N = 86 \cdot N = 87 \cdot N = 88 \cdot N = 89 \cdot N = 90 \cdot N = 91 \cdot N = 92 \cdot N = 93 \cdot N = 94 \cdot N = 95 \cdot N = 96 \cdot N = 97 \cdot N = 98 \cdot N = 99 \cdot N = 100 \cdot N = 105 \cdot N = 120$

**006. Packings of lines in  $\mathbb{R}^{15}$ .** First values for 50 blank cells ( $N = 51-100$ ) in the row  $m = 15, n = 1$  of the table of best Grassmannian packings, blank in Sloane's 1997 table and in Cohn's current one. Each value beats every packing obtained by deleting subspaces from a larger known one.

$N = 51 \cdot N = 52 \cdot N = 53 \cdot N = 54 \cdot N = 55 \cdot N = 56 \cdot N = 57 \cdot N = 58 \cdot N = 59 \cdot N = 60 \cdot N = 61 \cdot N = 62 \cdot N = 63 \cdot N = 64 \cdot N = 65 \cdot N = 66 \cdot N = 67 \cdot N = 68 \cdot N = 69 \cdot N = 70 \cdot N = 71 \cdot N = 72 \cdot N = 73 \cdot N = 74 \cdot N = 75 \cdot N = 76 \cdot N = 77 \cdot N = 78 \cdot N = 79 \cdot N = 80 \cdot N = 81 \cdot N = 82 \cdot N = 83 \cdot N = 84 \cdot N = 85 \cdot N = 86 \cdot N = 87 \cdot N = 88 \cdot N = 89 \cdot N = 90 \cdot N = 91 \cdot N = 92 \cdot N = 93 \cdot N = 94 \cdot N = 95 \cdot N = 96 \cdot N = 97 \cdot N = 98 \cdot N = 99 \cdot N = 100$

**007. Packings of lines in  $\mathbb{R}^{16}$ .** First values for 8 blank cells ( $N = 51-58$ ) in the row  $m = 16, n = 1$  of the table of best Grassmannian packings, blank in Sloane's 1997 table and in Cohn's current one. Each value beats every packing obtained by deleting subspaces from a larger known one.

$N = 51 \cdot N = 52 \cdot N = 53 \cdot N = 54 \cdot N = 55 \cdot N = 56 \cdot N = 57 \cdot N = 58$

# Coding theory

**008. Packings of planes in  $\mathbb{R}^8$ .** First values for 10 blank cells ( $N = 102, 104, 106, 108-109, 111, 113, 117, 119-120$ ) in the row  $m = 8, n = 2$  of the table of best Grassmannian packings, blank in Sloane's 1997 table and in Cohn's current one. Each value beats every packing obtained by deleting subspaces from a larger known one.

$N = 102 \cdot N = 104 \cdot N = 106 \cdot N = 108 \cdot N = 109 \cdot N = 111 \cdot N = 113 \cdot N = 117 \cdot N = 119 \cdot N = 120$

**009. Packings of planes in  $\mathbb{R}^{11}$ .** First values for 17 blank cells ( $N = 84-100$ ) in the row  $m = 11, n = 2$  of the table of best Grassmannian packings, blank in Sloane's 1997 table and in Cohn's current one. Each value beats every packing obtained by deleting subspaces from a larger known one.

$N = 84 \cdot N = 85 \cdot N = 86 \cdot N = 87 \cdot N = 88 \cdot N = 89 \cdot N = 90 \cdot N = 91 \cdot N = 92 \cdot N = 93 \cdot N = 94 \cdot N = 95 \cdot N = 96 \cdot N = 97 \cdot N = 98 \cdot N = 99 \cdot N = 100$

**010. Packings of planes in  $\mathbb{R}^{12}$ .** First values for 6 blank cells ( $N = 95-100$ ) in the row  $m = 12, n = 2$  of the table of best Grassmannian packings, blank in Sloane's 1997 table and in Cohn's current one. Each value beats every packing obtained by deleting subspaces from a larger known one.

$N = 95 \cdot N = 96 \cdot N = 97 \cdot N = 98 \cdot N = 99 \cdot N = 100$

**011. Packings of planes in  $\mathbb{R}^{13}$ .** First values for 46 blank cells ( $N = 51, 53\text{--}88, 92\text{--}100$ ) in the row  $m = 13, n = 2$  of the table of best Grassmannian packings, blank in Sloane's 1997 table and in Cohn's current one. Each value beats every packing obtained by deleting subspaces from a larger known one.

N=51 · N=53 · N=54 · N=55 · N=56 · N=57 · N=58 · N=59 · N=60 · N=61 · N=62 · N=63 · N=64 · N=65 ·  
N=66 · N=67 · N=68 · N=69 · N=70 · N=71 · N=72 · N=73 · N=74 · N=75 · N=76 · N=77 · N=78 · N=79 ·  
N=80 · N=81 · N=82 · N=83 · N=84 · N=85 · N=86 · N=87 · N=88 · N=92 · N=93 · N=94 · N=95 · N=96 ·  
N=97 · N=98 · N=99 · N=100

012. **Packings of planes in  $\mathbb{R}^{14}$ .** First values for 49 blank cells ( $N = 51\text{--}55, 57\text{--}100$ ) in the row  $m = 14$ ,  $n = 2$  of the table of best Grassmannian packings, blank in Sloane's 1997 table and in Cohn's current one. Each value beats every packing obtained by deleting subspaces from a larger known one.

N=51 · N=52 · N=53 · N=54 · N=55 · N=57 · N=58 · N=59 · N=60 · N=61 · N=62 · N=63 · N=64 · N=65 ·  
N=66 · N=67 · N=68 · N=69 · N=70 · N=71 · N=72 · N=73 · N=74 · N=75 · N=76 · N=77 · N=78 · N=79 ·  
N=80 · N=81 · N=82 · N=83 · N=84 · N=85 · N=86 · N=87 · N=88 · N=89 · N=90 · N=91 · N=92 · N=93 ·  
N=94 · N=95 · N=96 · N=97 · N=98 · N=99 · N=100

013. **Packings of planes in  $\mathbb{R}^{15}$ .** First values for 49 blank cells ( $N = 51\text{--}59, 61\text{--}100$ ) in the row  $m = 15$ ,  $n = 2$  of the table of best Grassmannian packings, blank in Sloane's 1997 table and in Cohn's current one. Each value beats every packing obtained by deleting subspaces from a larger known one.

N=51 · N=52 · N=53 · N=54 · N=55 · N=56 · N=57 · N=58 · N=59 · N=61 · N=62 · N=63 · N=64 · N=65 ·  
N=66 · N=67 · N=68 · N=69 · N=70 · N=71 · N=72 · N=73 · N=74 · N=75 · N=76 · N=77 · N=78 · N=79 ·  
N=80 · N=81 · N=82 · N=83 · N=84 · N=85 · N=86 · N=87 · N=88 · N=89 · N=90 · N=91 · N=92 · N=93 ·  
N=94 · N=95 · N=96 · N=97 · N=98 · N=99 · N=100

**014. Packings of three-spaces in  $\mathbb{R}^8$ .** First values for 13 blank cells (N = 101–102, 104–105, 107, 109–110, 112, 114, 116, 118–120) in the row m = 8, n = 3 of the table of best Grassmannian packings, blank in Sloane’s 1997 table and in Cohn’s current one. Each value beats every packing obtained by deleting subspaces from a larger known one.

N=101 · N=102 · N=104 · N=105 · N=107 · N=109 · N=110 · N=112 · N=114 · N=116 · N=118 · N=119 · N=120

**015. Packings of three-spaces in  $\mathbb{R}^{11}$ .** First values for 17 blank cells ( $N = 84\text{--}100$ ) in the row  $m = 11$ ,  $n = 3$  of the table of best Grassmannian packings, blank in Sloane's 1997 table and in Cohn's current one. Each value beats every packing obtained by deleting subspaces from a larger known one.

[illegible]

**016. Packings of three-spaces in  $\mathbb{R}^{12}$ .** First values for 6 blank cells ( $N = 95$ – $100$ ) in the row  $m = 12$ ,  $n = 3$  of the table of best Grassmannian packings, blank in Sloane's 1997 table and in Cohn's current one. Each value beats every packing obtained by deleting subspaces from a larger known one.

N=95 · N=96 · N=97 · N=98 · N=99 · N=100

017. **Packings of three-spaces in  $\mathbb{R}^{13}$ .** First values for 45 blank cells ( $N = 51, 53\text{--}64, 66\text{--}88, 92\text{--}100$ ) in the row  $m = 13, n = 3$  of the table of best Grassmannian packings, blank in Sloane's 1997 table and in Cohn's current one. Each value beats every packing obtained by deleting subspaces from a larger known one.

N=51 · N=53 · N=54 · N=55 · N=56 · N=57 · N=58 · N=59 · N=60 · N=61 · N=62 · N=63 · N=64 · N=66 ·  
N=67 · N=68 · N=69 · N=70 · N=71 · N=72 · N=73 · N=74 · N=75 · N=76 · N=77 · N=78 · N=79 · N=80 ·  
N=81 · N=82 · N=83 · N=84 · N=85 · N=86 · N=87 · N=88 · N=92 · N=93 · N=94 · N=95 · N=96 · N=97 ·  
N=98 · N=99 · N=100

**018. Packings of three-spaces in  $\mathbb{R}^{14}$ .** First values for 48 blank cells ( $N = 51-55, 57-69, 71-100$ ) in the row  $m = 14, n = 3$  of the table of best Grassmannian packings, blank in Sloane's 1997 table and in Cohn's current one. Each value beats every packing obtained by deleting subspaces from a larger known one.

N=51 · N=52 · N=53 · N=54 · N=55 · N=57 · N=58 · N=59 · N=60 · N=61 · N=62 · N=63 · N=64 · N=65 ·  
N=66 · N=67 · N=68 · N=69 · N=71 · N=72 · N=73 · N=74 · N=75 · N=76 · N=77 · N=78 · N=79 · N=80 ·  
N=81 · N=82 · N=83 · N=84 · N=85 · N=86 · N=87 · N=88 · N=89 · N=90 · N=91 · N=92 · N=93 · N=94 ·  
N=95 · N=96 · N=97 · N=98 · N=99 · N=100

**019. Packings of three-spaces in  $\mathbb{R}^{15}$ .** First values for 46 blank cells ( $N = 51-58, 61-73, 76-100$ ) in the row  $m = 15, n = 3$  of the table of best Grassmannian packings, blank in Sloane's 1997 table and in Cohn's current one. Each value beats every packing obtained by deleting subspaces from a larger known one.

N=51 · N=52 · N=53 · N=54 · N=55 · N=56 · N=57 · N=58 · N=61 · N=62 · N=63 · N=64 · N=65 · N=66 ·  
N=67 · N=68 · N=69 · N=70 · N=71 · N=72 · N=73 · N=76 · N=77 · N=78 · N=79 · N=80 · N=81 · N=82 ·  
N=83 · N=84 · N=85 · N=86 · N=87 · N=88 · N=89 · N=90 · N=91 · N=92 · N=93 · N=94 · N=95 · N=96 ·  
N=97 · N=98 · N=99 · N=100

**020. Packings of three-spaces in  $\mathbb{R}^{16}$ .** First values for 49 blank cells ( $N = 51$ – $62$ ,  $65$ – $78$ ,  $81$ – $100$ ,  $104$ ,  $107$ ,  $120$ ) in the row  $m = 16$ ,  $n = 3$  of the table of best Grassmannian packings, blank in Sloane's 1997 table and in Cohn's current one. Each value beats every packing obtained by deleting subspaces from a larger known one.

$N = 51 \cdot N = 52 \cdot N = 53 \cdot N = 54 \cdot N = 55 \cdot N = 56 \cdot N = 57 \cdot N = 58 \cdot N = 59 \cdot N = 60 \cdot N = 61 \cdot N = 62 \cdot N = 65 \cdot N = 66 \cdot$   
 $N = 67 \cdot N = 68 \cdot N = 69 \cdot N = 70 \cdot N = 71 \cdot N = 72 \cdot N = 73 \cdot N = 74 \cdot N = 75 \cdot N = 76 \cdot N = 77 \cdot N = 78 \cdot N = 81 \cdot N = 82 \cdot$   
 $N = 83 \cdot N = 84 \cdot N = 85 \cdot N = 86 \cdot N = 87 \cdot N = 88 \cdot N = 89 \cdot N = 90 \cdot N = 91 \cdot N = 92 \cdot N = 93 \cdot N = 94 \cdot N = 95 \cdot N = 96 \cdot$   
 $N = 97 \cdot N = 98 \cdot N = 99 \cdot N = 100 \cdot N = 104 \cdot N = 107 \cdot N = 120$

**021. Packings of four-spaces in  $\mathbb{R}^9$ .** First values for 15 blank cells ( $N = 5$ – $19$ ) in the row  $m = 9$ ,  $n = 4$  of the table of best Grassmannian packings, blank in Sloane's 1997 table and in Cohn's current one. Each value beats every packing obtained by deleting subspaces from a larger known one.

$N = 5 \cdot N = 6 \cdot N = 7 \cdot N = 8 \cdot N = 9 \cdot N = 10 \cdot N = 11 \cdot N = 12 \cdot N = 13 \cdot N = 14 \cdot N = 15 \cdot N = 16 \cdot N = 17 \cdot N = 18 \cdot N = 19$

**022. Packings of four-spaces in  $\mathbb{R}^{10}$ .** First values for 15 blank cells ( $N = 6$ – $20$ ) in the row  $m = 10$ ,  $n = 4$  of the table of best Grassmannian packings, blank in Sloane's 1997 table and in Cohn's current one. Each value beats every packing obtained by deleting subspaces from a larger known one.

$N = 6 \cdot N = 7 \cdot N = 8 \cdot N = 9 \cdot N = 10 \cdot N = 11 \cdot N = 12 \cdot N = 13 \cdot N = 14 \cdot N = 15 \cdot N = 16 \cdot N = 17 \cdot N = 18 \cdot N = 19 \cdot$   
 $N = 20$

**023. Packings of four-spaces in  $\mathbb{R}^{11}$ .** First values for 34 blank cells ( $N = 5$ – $20$ ,  $83$ – $100$ ) in the row  $m = 11$ ,  $n = 4$  of the table of best Grassmannian packings, blank in Sloane's 1997 table and in Cohn's current one. Each value beats every packing obtained by deleting subspaces from a larger known one.

$N = 5 \cdot N = 6 \cdot N = 7 \cdot N = 8 \cdot N = 9 \cdot N = 10 \cdot N = 11 \cdot N = 12 \cdot N = 13 \cdot N = 14 \cdot N = 15 \cdot N = 16 \cdot N = 17 \cdot N = 18 \cdot$   
 $N = 19 \cdot N = 20 \cdot N = 83 \cdot N = 84 \cdot N = 85 \cdot N = 86 \cdot N = 87 \cdot N = 88 \cdot N = 89 \cdot N = 90 \cdot N = 91 \cdot N = 92 \cdot N = 93 \cdot N = 94 \cdot$   
 $N = 95 \cdot N = 96 \cdot N = 97 \cdot N = 98 \cdot N = 99 \cdot N = 100$

**024. Packings of four-spaces in  $\mathbb{R}^{12}$ .** First values for 21 blank cells ( $N = 5$ – $20$ ,  $94$ – $95$ ,  $97$ ,  $99$ – $100$ ) in the row  $m = 12$ ,  $n = 4$  of the table of best Grassmannian packings, blank in Sloane's 1997 table and in Cohn's current one. Each value beats every packing obtained by deleting subspaces from a larger known one.

$N = 5 \cdot N = 6 \cdot N = 7 \cdot N = 8 \cdot N = 9 \cdot N = 10 \cdot N = 11 \cdot N = 12 \cdot N = 13 \cdot N = 14 \cdot N = 15 \cdot N = 16 \cdot N = 17 \cdot N = 18 \cdot$   
 $N = 19 \cdot N = 20 \cdot N = 94 \cdot N = 95 \cdot N = 97 \cdot N = 99 \cdot N = 100$

**025. Packings of four-spaces in  $\mathbb{R}^{13}$ .** First values for 32 blank cells ( $N = 5$ – $27$ ,  $92$ – $100$ ) in the row  $m = 13$ ,  $n = 4$  of the table of best Grassmannian packings, blank in Sloane's 1997 table and in Cohn's current one. Each value beats every packing obtained by deleting subspaces from a larger known one.

$N = 5 \cdot N = 6 \cdot N = 7 \cdot N = 8 \cdot N = 9 \cdot N = 10 \cdot N = 11 \cdot N = 12 \cdot N = 13 \cdot N = 14 \cdot N = 15 \cdot N = 16 \cdot N = 17 \cdot N = 18 \cdot$   
 $N = 19 \cdot N = 20 \cdot N = 21 \cdot N = 22 \cdot N = 23 \cdot N = 24 \cdot N = 25 \cdot N = 26 \cdot N = 27 \cdot N = 92 \cdot N = 93 \cdot N = 94 \cdot N = 95 \cdot N = 96 \cdot$   
 $N = 97 \cdot N = 98 \cdot N = 99 \cdot N = 100$

## Discrete geometry

**026. Spreading points in the plane.** First values for  $n = 32$ – $50$  (19 cells) in Erich Friedman's table of point sets in the plane minimizing the ratio of the largest to the smallest distance, which stops at  $n = 31$ . Each value is smaller than what deleting a point from the next configuration gives.

$n = 32 \cdot n = 33 \cdot n = 34 \cdot n = 35 \cdot n = 36 \cdot n = 37 \cdot n = 38 \cdot n = 39 \cdot n = 40 \cdot n = 41 \cdot n = 42 \cdot n = 43 \cdot n = 44 \cdot n = 45 \cdot n = 46 \cdot$   
 $n = 47 \cdot n = 48 \cdot n = 49 \cdot n = 50$